

KHYAGT, Russian FederationWMO: **309250**Lat: **50.364N**Lon: **106.452E**Elev: **797**StdP: **92.11**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-28.8	-33.4	0.2	-30.5	-31.5	0.2	-28.5	7.1	-15.0	5.5	-13.9	0.2	20	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.5	18.8	28.5	17.9	26.6	17.2	20.3	28.1	19.0	26.3	18.0	24.7	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	13.8	24.0	16.4	12.9	22.5	15.4	12.0	21.6	62.5	28.1	57.8	26.5	54.4	24.7	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.4	5.4	-32.7	34.1	3.0	2.4	-34.8	35.8	-36.6	37.3	-38.3	38.6	-40.5	40.4		
(5)			-32.7	21.9	3.0	1.5	-34.8	23.0	-36.6	23.8	-38.3	24.7	-40.5	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-19.8	-15.3	-5.7	3.9	10.5	17.3	19.9	17.3	10.3	1.5	-9.0	-17.4	(6)
(7)		DBStd	14.61	5.00	5.76	6.26	5.07	5.09	4.30	3.48	3.67	4.48	5.02	6.21	5.35	(7)
(8)		HDD10.0	4106	923	709	487	192	57	2	0	1	50	266	569	850	(8)
(9)		HDD18.3	6395	1181	942	745	434	247	69	22	64	241	522	819	1108	(9)
(10)		CDD10.0	898	0	0	0	8	73	222	308	226	60	2	0	0	(10)
(11)		CDD18.3	147	0	0	0	0	4	39	71	31	2	0	0	0	(11)
(12)		CDH23.3	1369	0	0	0	2	78	398	628	251	13	0	0	0	(12)
(13)		CDH26.7	390	0	0	0	0	16	113	199	61	1	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.0	1.3	2.0	2.6	2.8	2.4	2.0	1.9	2.0	1.8	1.5	1.0	(14)
(15)	Precipitation	PrecAvg	350	4	4	6	14	20	66	87	79	42	14	7	4	(15)
(16)		PrecMax	617	14	10	20	58	48	191	212	231	78	41	19	12	(16)
(17)		PrecMin	216	0	0	0	1	6	8	17	27	9	1	1	1	(17)
(18)		PrecStd	86	3	3	5	13	12	40	46	44	21	11	5	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.0	0.5	12.7	21.4	29.3	32.5	33.9	31.3	25.0	17.0	7.1	-3.3	(19)
(20)			MCWB	-6.8	-2.9	5.2	10.5	15.6	19.1	21.6	20.2	16.0	9.3	2.3	-5.3	(20)
(21)		2%	DB	-8.2	-2.7	8.3	18.0	25.6	29.5	31.1	28.2	22.2	13.7	4.1	-6.3	(21)
(22)			MCWB	-9.7	-5.0	2.8	8.7	14.0	17.5	19.6	18.6	14.4	7.0	0.4	-7.9	(22)
(23)		5%	DB	-10.8	-5.0	5.6	15.2	23.0	27.8	29.0	26.1	20.0	11.1	1.9	-8.3	(23)
(24)			MCWB	-12.0	-6.8	1.0	7.1	12.6	16.8	18.8	17.5	12.9	5.6	-0.9	-9.6	(24)
(25)		10%	DB	-12.9	-7.1	3.3	12.7	20.3	25.5	26.9	24.0	17.7	8.6	-0.3	-10.1	(25)
(26)			MCWB	-13.8	-8.5	-0.6	5.7	11.1	16.0	18.0	16.5	11.7	4.1	-2.5	-11.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.8	-2.7	5.5	11.1	16.6	20.5	22.8	21.1	17.2	9.4	2.6	-5.4	(27)
(28)			MCDWB	-5.3	0.1	12.4	20.2	27.8	29.8	32.1	29.4	23.8	16.3	6.6	-3.5	(28)
(29)		2%	WB	-9.6	-5.0	3.0	9.1	14.6	18.8	21.0	19.3	15.1	7.4	0.7	-7.8	(29)
(30)			MCDWB	-8.3	-2.8	8.2	17.1	24.4	27.4	29.0	26.7	20.6	13.3	3.8	-6.3	(30)
(31)		5%	WB	-12.0	-6.8	1.2	7.4	13.1	17.7	19.7	18.1	13.6	5.8	-0.9	-9.5	(31)
(32)			MCDWB	-10.9	-5.1	5.4	14.7	22.2	25.6	26.7	24.5	18.9	10.6	1.9	-8.3	(32)
(33)		10%	WB	-13.8	-8.5	-0.5	5.8	11.5	16.7	18.6	17.0	12.2	4.3	-2.4	-11.1	(33)
(34)			MCDWB	-12.9	-7.1	3.1	12.4	19.6	24.0	25.3	22.8	17.2	8.3	-0.3	-10.1	(34)
(35)	Mean Daily Temperature Range		MDBR	7.4	9.4	10.7	11.3	12.2	11.2	10.0	9.8	10.5	9.8	8.4	7.3	(35)
(36)		5% DB	MCDBR	8.2	10.5	12.3	15.2	16.2	14.6	13.2	12.9	13.2	13.0	9.8	8.1	(36)
(37)			MCWBR	7.2	8.5	8.1	9.0	8.9	6.8	5.6	6.0	6.8	7.8	6.8	6.9	(37)
(38)		5% WB	MCDWB	8.1	10.3	11.9	14.5	15.3	12.5	11.2	11.6	11.9	12.4	9.5	8.1	(38)
(39)			MCWBR	7.1	8.4	7.9	8.9	8.7	6.4	5.4	5.8	6.3	7.6	6.8	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.279	0.317	0.363	0.412	0.414	0.403	0.372	0.323	0.297	0.276	0.259	(40)	
(41)		taud	2.252	2.237	2.262	2.250	2.130	2.237	2.313	2.382	2.468	2.467	2.340	2.227	(41)	
(42)		Ebn at Noon	784	858	885	878	844	845	848	860	875	838	755	714	(42)	
(43)		Edh at Noon	69	92	109	124	146	133	121	108	88	73	63	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.58	2.76	4.17	5.08	6.05	6.26	5.79	4.95	3.96	2.58	1.59	1.18	(44)	
(45)		RadStd	0.05	0.08	0.20	0.26	0.34	0.30	0.34	0.31	0.32	0.16	0.07	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page